

INFLUENCE OF HYDROCARBON CHAIN LENGTH ON CRITICAL MICELLE CONCENTRATION OF VARIOUS SURFACTANTS

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ABSTRACT:

Critical Micelle Concentration (CMCs) of various surfactants (cationic, anionic and non-ionic) have been determined in carbonate-bicarbonate buffer at 35°C by different techniques such as surface tension (S.T.) conductivity and dye solubilization.

It has been found that in all cases CMC's of surfactants decrease with increasing hydrophobic part of the surfactants.

INTRODUCTION

One of the most outstanding properties of the surface active agents is their tendency of self association leading to micelle formation. Micellae formation is important in many chemical and biological processes (Magid L. 1979).

The presence of micelle is reflected by the CMC which is effected by various factors like structure of the surfactant, organic additives in solution and temperature.

Critical micelle concentration of surfactant is a significant parameter to understand the phenomenon of micellization of the surfactant. The value of the CMC depends on a large number of parameters usually the more surface active the amphiphilic monomer, the higher is the tendency for micellization and hence the lower the CMC of the micelle produced. According by the longer the total carbon chain length of the monomeric surfactant, the lower the CMC becomes. The number of carbon atoms, n is related to the logarithm of the CMC.

$$\log \text{CMC} = A - Bn \quad (1)$$

where A and B are constants.

In the present work the influence of hydrocarbon chain length on the CMCs is determined by S.T. conductivity and dye solubilization.

MATERIALS AND METHODS

Surfactants:

With the exception of the following surfactants, all other surfactants studied in this course of work were used as such without subjecting to purification process.

Anionic-Surface-Active Agents:

Sodium dodecyl sulphate (SDS) ($C_{12}H_{25}NaSO_4$): The A.R. Grade of sodium dodecyl sulphate gave a melting point $210^{\circ}C$ (Lit. $204-207^{\circ}C$) (Aldrich Catalogue).

The major impurities were supposed to be inorganic salts homologous series of the particular substance, and long chain sulphate, unesterified alcohol remaining from the preparation. The sample was purified by successive treatments of diethyl ether followed by repeated crystallization from water as described by William et al. (1955) and Dreger et al. (1944). The purified sample of SDS gave a melting point $205^{\circ}C$.

Cationic Surface-Active Agents:

Dodecyl trimethyl ammonium bromide ($C_{12}H_{25}N^{+}(CH_3)_3Br^{-}$): It was purified by recrystallization once from benzene-ether and from acetone-ether. The product decomposed at $280^{\circ}C$ (Lit. 207°) (Bruneng et al., 1961).

Tetradecyl trimethyl ammonium bromide ($C_{14}H_{29}N^{+}(CH_3)_3Br^{-}$): The surfactant was purified by repeated crystallization of alcohol until there was no minimum in the surface tension log concentration profile (Venalue et al., 1964).

Hexadecyl trimethyl ammonium bromide ($C_{16}H_{31}N^{+}(CH_3)_3Br^{-}$): The reagent grade sample of CTAB (Sigma) was subjected to the purification process described by Mukerjee and Mysels (1955) and Dystee and Grunwald (1959).

Water: Freshly distilled water from an all glass still was used which has specific conductivity $< 10^{-7} \text{ ohm}^{-1} \text{ cm}^{-1}$ and surface tension of 72.05 mNm^{-1} at $25^{\circ}C$.

Buffer solution: Delory and King's carbonate-bicarbonate buffer at pH 9.2 was prepared according to Documenta Gagy (1962).

Tensiometer: Model 21, Cat. No.14-814, supplied by Fischer Scientific.

Conductivity meter: Supplied by Karl Kob, Scientific Technical Suppliers.

EXPERIMENTAL

Measurement of Critical Micelle Concentration (CMC)

The usefulness of CMC values in various quantitative and qualitative investigations involving surfactant solution arises basically from the fact that the surface and the interfacial activity of the (polar-nonpolar) monomers is closely reflected in the value of the CMC. The methods available for CMC measurements have been reviewed by Shinoda et al. (1963). Mukerjee and Mysels (1971) have reported about 71 methods of determining the CMC ranging from a few widely used once to a few reported only once. Virtually any physical property which varies according to whether the solute is monomeric or aggregated can be used to determine the CMC, by measuring its variation with concentration of surfactants. For this study the CMC of various ionic and nonionic surfactants at temperature 35°C were measured by conductivity, surface tension and dye solubilization method.

1. Surface Tension

Surface tension measurements have been used by numerous workers in order to determine CMCs. The method has a great advantage in that it can be applied to ionic and nonionic surfactants and can be used down to very low concentrations. Various techniques available to measure surface tension (Harkins et al., 1966) of which the Whelmy slide method, in which force is required to detach a thin microscopic slide of known thickness and width from the surface of a liquid is measured, has been used here.

The detailed operating procedure and a diagrammatic representation of process are given by Harkins and Alexander (1960) and Shetewi (1975).

Surface tension of various concentration of surfactant were then plotted against log concentration of surfactants and the CMC determined from the intersection of the 2 linear parts. A typical set of data is shown in Table 1 and is plotted in Fig.1. Other CMC values obtained by this method are shown elsewhere (Hassan F. 1996).

2. Conductivity Method:

The apparent critical micelle concentration (CMC) was measured by using conductivity meter. In the present study this method of CMC measurement was valid for temperature 40°C only since the range of the instrument for the temperature was between -5° to 50°C.

Great care was taken to clean the electrode after each determination by ethanol rinsed repeatedly with deionized water and allow to dry. Pure water was used all the

time to check the validity of the method and the accuracy of the instrument. A typical set of data obtained by this method at 40°C has been given in Table 1.1 and is plotted in Fig. 1.1. Other CMC values obtained by this technique are given elsewhere (Hassan F. 1996).

3. Dye Solubilization Method:

The CMCs of various ionic and nonionic surfactants were determined by dye solubilization method. using 4-dimethyl aminobenzol dye, a sharp change in the slope of a plot of optical density versus molar concentration indicates the CMC. For the determination of CMCs, a series of concentration of surfactants were prepared in buffer solutions 25 ml of each concentrations were taken in 100 ml volumetric flask or in a dispensing bottle. Small quantity of dye was added to each flask. These solutions were shaken thoroughly, on a temperature controlled shaking water bath for 30 minutes. The solutions were filtered with No.1.5 mm filter paper in a beaker 5 ml of each filtrate solution was taken in 25 ml volumetric flask and the volume was make up by alcohol. The optical density of each concentration was recorded on spectrophotometer at 413 nm. Absorbance of various concentrations of surfactants were then plotted against molar concentrations of each surfactant and CMC determined from the intersection of the 2 linear parts. A typical set of data has been tabulated in Table 1.2 and plotted in Fig. 1.1. Other CMC values obtained by this method are given elsewhere (Hassan F. 1996).

DISCUSSION

The data obtained in this study were plotted according to equation (1) and a typical set of plots for anionic, cationic and nonionic surfactants is shown in Fig. 1.3 constant B of equation (1) was calculated for this study. The value obtained closely resemble the published values which are ranged between 0.2-0.5 (Molynex et al., 1965; Clint et al., 1975; Lin et al., 1971).

The values of CMCs of surfactants used in this study have been reported in Table 1.3-1.5. It is apparent from the Table 1.3-1.5 that increase in the number of carbon atoms in unbranched hydrocarbon chain leads to a decrease in the CMC (Mukerjee et al., 1971; Attwood et al., 1983). The phenomenon of micelle formation in aqueous system is primarily a result of hydrophobic interaction. The polar heads of surfactants interact strongly with water molecules while the interaction between water molecule and the nonpolar part of the surfactant is highly unfavourable, there is a tendency for the nonpolar tails to come in contact with each other. This interaction between nonpolar parts due to unfavourable conditions to interact with water molecules is known as 'hydrophobic bonding'. In a given homologous series of normal hydrocarbon chain surfactants for longer chain length micellization is more probable i.e. longer the

chain, the larger is the portion of the nonpolar part involved in hydrophobic bonding, since hydrophobic bonding has a strong additive character substantial free energy changes are involved in micelle formation. The decrease of interfacial energy on micellization is more for longer chain than for short ones and it is possible that there is more water loss from transfer of a long chain into micelles than a short one.

Table 1.0
Replicate values of surface tension of CTAB in carbonate-Bicarbonate buffer
at pH 9.2 and 35°C
(Wilhelmy Plate Technique)

Molar concentration of CTAB	Surface tension (mNm ⁻¹)			Mean
	I	II	III	
1 x 10 ⁻⁵	43.5	43.5	43.0	43.33
5 x 10 ⁻⁵	42.0	42.5	41.5	42.00
1 x 10 ⁻⁴	41.5	42.0	41.0	41.50
5 x 10 ⁻⁴	41.5	41.0	40.0	40.83
1 x 10 ⁻³	39.5	40.0	39.5	39.66
3 x 10 ⁻³	39.5	39.0	38.5	39.00
5 x 10 ⁻³	39.0	38.5	39.0	38.83
8 x 10 ⁻³	39.5	39.0	38.5	39.00
1 x 10 ⁻²	38.5	38.5	38.5	38.50
3 x 10 ⁻²	38.5	39.0	39.0	38.83
5 x 10 ⁻²	38.5	39.0	38.5	38.66
8 x 10 ⁻²	37.0	37.5	37.5	37.33

Table 1.1
Replicate values of surface tension of SDS in carbonate-Bicarbonate buffer
at pH 9.2 and 35°C
(Conductivity Method)

Molar concentration of SDS	Conductivity values (ms/cm)			Mean
	I	II	III	
1×10^{-5}	7.85	7.80	7.88	7.84
5×10^{-5}	7.90	7.88	7.90	7.89
1×10^{-4}	7.91	7.90	7.94	7.92
5×10^{-4}	7.95	7.94	7.93	7.94
1×10^{-3}	7.95	7.93	7.94	7.94
3×10^{-3}	7.99	7.98	7.95	7.97
5×10^{-3}	8.00	8.00	7.98	7.99
8×10^{-3}	8.30	8.35	8.30	8.32
1×10^{-2}	8.60	8.57	8.62	8.60
3×10^{-2}	8.80	8.90	8.90	8.87
5×10^{-2}	9.20	9.18	9.24	9.21
8×10^{-2}	9.60	9.60	9.58	9.59

Table 1.2
Replicate values of surface tension of tween 80 in
Carbonate-Bicarbonate buffer at pH 9.2 and 35°C
(Dye Solubilization Method)

% age concentration of Tween 80 w/v	Absorbance at 413 nm			Mean
	I	II	III	
0.05	0.00	0.00	0.00	0.00
0.1	0.10	0.12	0.12	0.11
0.2	0.40	0.40	0.50	0.43
0.3	0.80	0.80	0.90	0.83
0.4	1.10	1.10	1.20	1.13
0.5	1.50	1.50	1.60	1.53
0.6	2.10	2.00	2.20	2.10
0.7	2.70	2.60	2.70	2.67
0.8	3.25	3.23	3.27	3.25

Table 1.3
CMC values of anionic surfactants measured at pH 9.2 and 70°C
under the conditions used for kinetic and physical studies

Anionic surfactants	System		Method	CMC
	Buffer	PNPP		
Sodium dodecyl sulphate	+	-	CM	4.0×10^{-3}
	+	+	ST(T)	4.0×10^{-3}
	+	+	ST(T)	4.0×10^{-3}
Sodium dodecyl sulphate	+	-	CM	3.0×10^{-3}
	+	-	ST(T)	3.0×10^{-3}
	+	+	Ds	3.0×10^{-3}
Sodium tetradecyl sulphate	+	-	ST(T)	2.0×10^{-3}
	+	-	Ds	2.0×10^{-3}
Sodium hexadecyl sulphate	+	-	ST(T)	2.5×10^{-5}
	+	+	ST(T)	2.5×10^{-5}
Sodium octadecyl sulphate	+	-	ST(T)	1.5×10^{-5}
			Ds	1.5×10^{-5}
	+	+	CM	1.5×10^{-5}

Key:

ST(T) = Surface tension by tensiometer

Ds = Dye solubilization

CM = Conductivity method

+ = Present

= Absent

Buffer = Carbonate-Bicarbonate

PNPP = p-nitrophenyl propionate

Table 1.4
CMC values of cationic surfactants measured at pH 9.2 and 30°C
under the conditions used for kinetic and physical studies

Cationic surfactants	System			Method	CMC
	Buffer	Salt	PNPP		
Dodecyl trimethyl ammonium bromide	+	-	-	ST(T)	5.0×10^{-3}
	+	-	+	ST(T)	5.0×10^{-3}
	+	-	-	CM	5.0×10^{-3}
Tetradecyl trimethyl ammonium bromide	+	-	-	ST(T)	3.5×10^{-3}
	+	-	+	ST(T)	3.5×10^{-3}
	+	-	-	CM	3.5×10^{-3}
Hexadecyl trimethyl ammonium bromide	+	+	-	ST(T)	3.0×10^{-3}
	+	-	+	ST(T)	3.0×10^{-3}
	+	-	-	CM	3.0×10^{-3}
Cetyl pyridinium bromide	+	-	-	ST(T)	3.0×10^{-4}
	+	-	+	ST(T)	3.0×10^{-4}
	+	-	-	CM	3.0×10^{-4}
	+	-	-	Ds	2.8×10^{-4}

Key:

ST(T) = Surface tension by tensiometer

Ds = Dye solubilization

CM = Conductivity method

+ = Present

- = Absent

Buffer = Carbonate-Bicarbonate

PNPP = p-nitrophenyl propionate

Table 1.5
CMC values of nonionic surfactants measured at pH 9.2 and 35°C
under conditions used for kinetic and physical studies

Surfactant	System			Method	CMC % w/v
	Buffer	Salt	PNPP		
Polysorbate (Tween) 20	+	-	-	ST(T)	1
	+	-	+	ST(T)	1
	+	-	-	Ds	0.9
	+	-	+	Ds	0.9
Polysorbate (Tween) 40	+	-	-	ST(T)	0.8
	+	-	+	ST(T)	0.8
	+	-	-	Ds	0.8
	+	-	+	Ds	0.8
Polysorbate (Tween) 60	+	-	-	ST(T)	0.5
	+	-	+	ST(T)	0.5
	+	-	-	Ds	0.49
	+	-	+	Ds	0.49
Polysorbate (Tween) 80	+	-	-	ST(T)	0.5
	+	-	+	ST(T)	0.5
	+	-	-	Ds	0.5
	+	-	+	Ds	0.5

Key:

ST(T) = Surface tension by tensiometer

Ds = Dye solubilization

CM = Conductivity method

+ = Present

- = Absent

Buffer = Carbonate-Bicarbonate

PNPP = p-nitrophenyl propionate

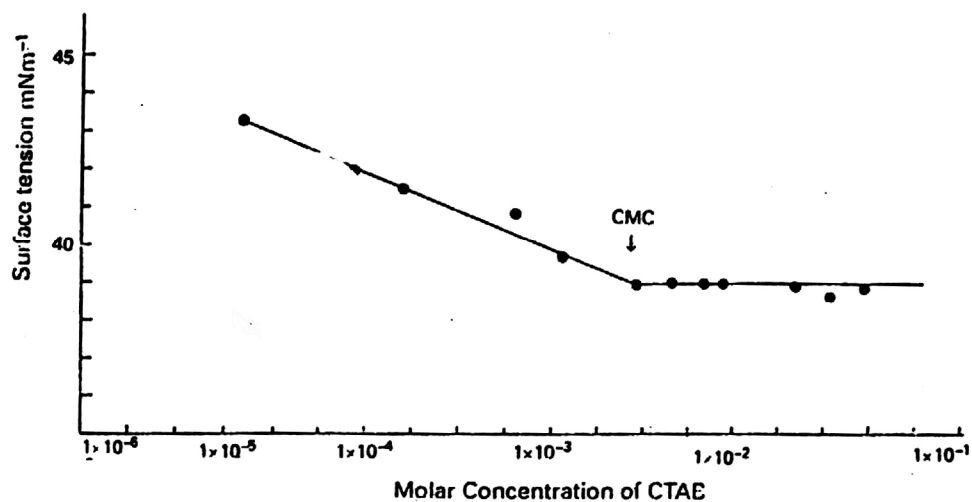


Figure 1.0: Plot of surface tension versus log concentration of CTAB in carbonate-bicarbonate buffer at pH 9.2 and 35°C.

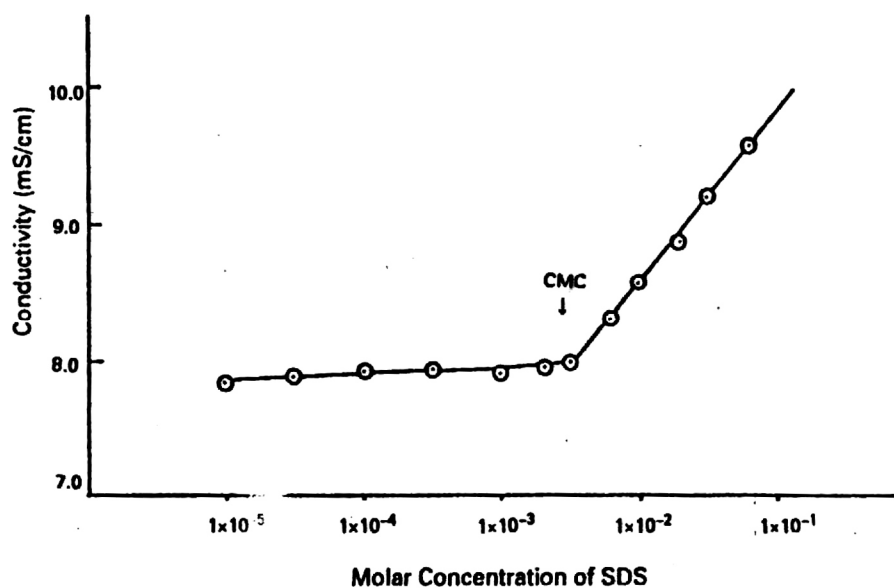


Figure 1.1: Plot of specific conductivity versus molar concentration of SDS in carbonate-bicarbonate buffer at pH 9.2 and 35°C.

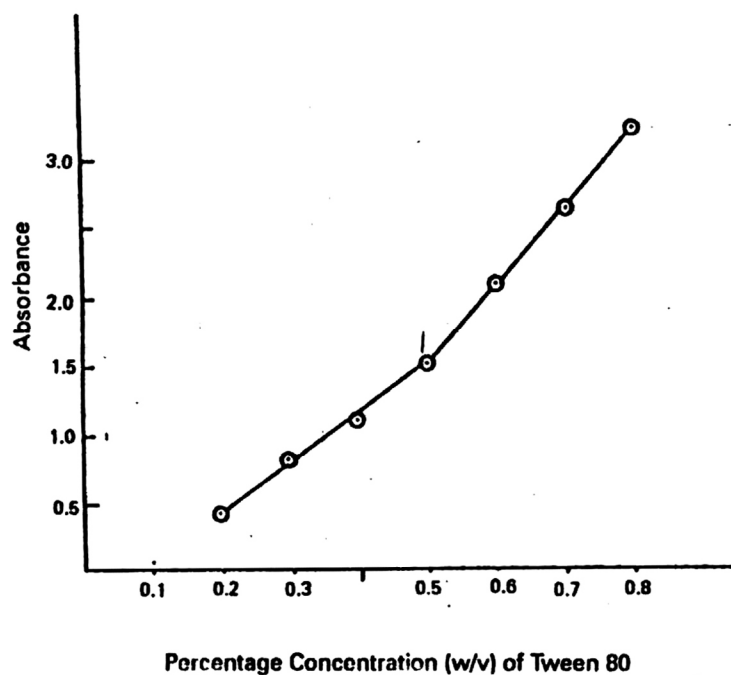


Figure 1.2: Plot of absorbance against percentage concentration of Tween 80 in carbonate-bicarbonate buffer at pH 9.2 and 35°C.

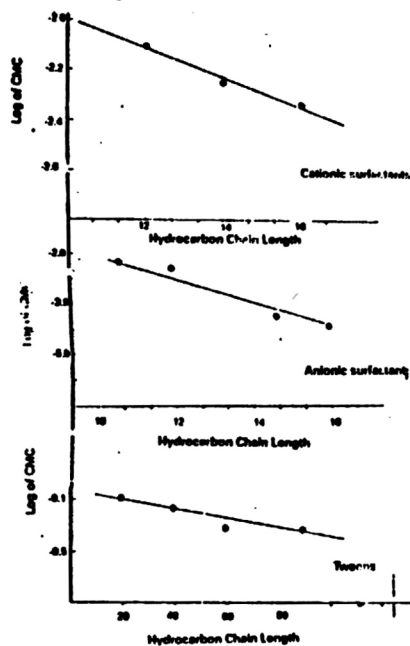


Figure 1.3: A plot of log CMC against hydrocarbon chain length in various surfactants.

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